Quantification: I534_z19c1-2
==========
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
---
measure	value
lacuna count	30
interior lacuna count	25
frame-edge lacuna count	5

Interior lacunae only: mean, median and sample SD (n = 25)
---
measure	kind	mean	median	sd	units
area_px2	lacuna shape	2807.24	1823	2158.83	px^2
major_axis_length_px	lacuna shape	84.1617	84.8432	29.7084	px
minor_axis_length_px	lacuna shape	45.368	38.4077	17.7887	px
aspect_ratio	lacuna shape	1.9499	1.9701	0.5721	unitless
eccentricity	lacuna shape	0.7942	0.8616	0.1582	unitless
solidity	lacuna shape	0.7506	0.7525	0.0958	unitless
roots_count	headline	10.8	9	4.3012	count
ring_length_r30_px	headline	275.88	267	89.6471	px
ring_length_r60_px	reported	516.48	526	157.111	px
owned_length_px	ownership-dependent	602.606	462.676	525.118	px
edge_count	ownership-dependent	29.6	21	24.1833	count
mean_edge_length_px	ownership-dependent	19.7499	19.7162	3.579	px
perimeter_px	normalised	287.499	283.463	98.2201	px
ring_area_r30_px2	normalised	9317	9124	2968.97	px^2
ring_area_r60_px2	normalised	18836.1	20421	5492.16	px^2
in_frame_fraction_r30	normalised	0.9975	1	0.0097	unitless
in_frame_fraction_r60	normalised	0.9827	1	0.0477	unitless
ring_density_r30	normalised	0.0312421	0.0297647	0.0111001	px^-1
ring_density_r60	normalised	0.0285877	0.0271714	0.00871518	px^-1
roots_per_100px_perimeter	normalised	3.9433	3.4593	1.4875	per 100 px
ring_attached_length_r30_px	network v2	246.72	238	83.0324	px
ring_attached_length_r60_px	network v2	374.44	390	135.282	px
ring_length_w_r30_px	network v2	313.466	296.995	102.851	px
ring_length_w_r60_px	network v2	586.987	597.6	179.934	px
sholl_crossings_r10	network v2	6.2	6	2.5331	count
sholl_crossings_r20	network v2	7.2	7	3.1754	count
sholl_crossings_r30	network v2	6.76	7	2.7731	count
loop_count	loops, report only	1.24	0	2.3324	count
loop_edge_share	loops, report only	0.0911	0	0.1139	share
ring_width_mean_r30_px	width	7.4163	7.4817	0.7073	px

Per field
---
measure	value	units
analysed_area_px2	953080	px^2
total_skeleton_length_px	23522	px
canalicular_length_density_per_px	0.02468	px^-1
junction_count	471	count
junction_density_per_px2	0.00049419	px^-2
lacuna_count	30	count

lacunae_per_px2	3.148e-05	px^-2
skeleton_component_count	338	count
mean_component_length_px	69.5917	px
median_component_length_px	24	px
field_length_density_w_per_px	0.028041	px^-1
field_density_without_flagged_per_px	0.0267402	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	470	px
unattached_component_count	28	count
canalicular_length_density_attached_per_px	0.02468	px^-1
field_length_density_with_unattached_per_px	0.0251731	px^-1
loop_count	40	count
loop_edge_share	0.1149	share
edge_count	1271	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.0822	px
width_median_px	8.2462	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	22537	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	7213	156.74	77.5309	2.0216	0.8691	0.6864	-1.375	29.9816	225.264	True
2	1583	53.2276	47.8198	1.1131	0.4392	0.7794	0.9204	26.8503	401.312	True
3	12484	376.172	54.3639	6.9195	0.9895	0.6519	1.5686	31.5111	841.265	True
4	2878	110.538	36.55	3.0243	0.9438	0.8432	-1.5019	47.2293	619.351	False
5	1304	60.2003	30.9259	1.9466	0.858	0.7893	-1.2145	90.1833	146.716	False
6	1823	56.4889	49.8735	1.1326	0.4696	0.8271	-1.1579	125.003	757.529	False
7	3354	87.3099	52.7612	1.6548	0.7968	0.8689	-1.2343	222.521	989.166	True
8	10008	137.275	97.443	1.4088	0.7044	0.8833	-1.0949	269.479	692.84	False
9	1637	57.8693	48.4179	1.1952	0.5477	0.7276	-0.0784	295.51	896.101	False
10	1291	47.4215	38.1837	1.2419	0.593	0.8741	0.4227	324.111	336.189	False
11	627	51.449	19.4729	2.6421	0.9256	0.7093	0.4876	337.973	69.1467	False
12	6026	129.945	61.3358	2.1186	0.8816	0.8485	-1.1185	377.349	511.866	False
13	1342	66.6735	36.4005	1.8317	0.8378	0.6286	1.399	396.13	787.283	False
14	1283	64.6772	37.6197	1.7192	0.8134	0.6386	1.1702	434.174	798.524	False
15	2978	75.2858	55.7926	1.3494	0.6714	0.7831	-0.8753	453.94	606.7	False
16	596	38.0384	25.7402	1.4778	0.7363	0.7469	-0.9961	464.5	212.371	False
17	4876	84.2608	78.5547	1.0726	0.3617	0.7525	0.8631	503.31	62.7231	False
18	3737	84.848	59.7745	1.4195	0.7097	0.8338	-0.4377	493.364	441.851	False
19	820	51.4532	24.5895	2.0925	0.8784	0.7614	1.0688	505.329	211.1	False
20	3553	104.824	44.321	2.3651	0.9062	0.8721	0.2384	581.822	856.943	False
21	4782	122.937	62.4014	1.9701	0.8616	0.6511	1.4416	592.132	630.596	False
22	1615	90.9802	31.7451	2.866	0.9372	0.5886	-0.9716	606.27	724.946	False
23	2632	101.535	38.4077	2.6436	0.9257	0.6745	-1.3449	587.391	245.703	False
24	5166	140.866	61.0892	2.3059	0.9011	0.6428	0.013	686.162	69.5337	False

25	1712	87.747	36.8261	2.3827	0.9077	0.6786	-0.5403	724.265	671.19	False
26	914	54.9878	33.7201	1.6307	0.7899	0.5745	-0.9811	788.83	430.015	False
27	2692	86.6013	40.7838	2.1234	0.8822	0.8628	-1.1084	820.32	99.4658	False
28	1680	84.8432	32.3344	2.6239	0.9245	0.7391	-0.8417	827.977	860.805	False
29	4209	112.297	51.8973	2.1638	0.8868	0.8336	-0.6033	855.455	275.629	False
30	681	43.231	24.4304	1.7696	0.825	0.8107	-1.479	862.358	1007.22	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	16	389	667	721.027	41	17.586	7.4086
2	3	152	432	242.338	14	17.3099	7.3565
3	7	249	474	167.51	13	12.8854	8.0593
4	6	167	343	91.7696	8	11.4712	8.4144
5	9	235	428	262.823	17	15.4602	6.1544
6	5	175	382	404.191	21	19.2472	7.2624
7	3	105	193	168.711	8	21.0888	6.6191
8	18	485	762	1393.5	72	19.3541	8.0435
9	7	132	249	259.936	14	18.5669	6.7288
10	9	269	718	2540.79	118	21.5321	6.7737
11	5	133	389	105.225	5	21.0451	5.3755
12	20	382	590	554.931	36	15.4148	7.0898
13	8	239	474	611.203	31	19.7162	7.5543
14	13	333	624	900.666	48	18.7639	7.1833
15	9	239	330	357.635	20	17.8817	8.132
16	10	244	428	509.002	35	14.5429	7.4817
17	16	421	621	834.323	36	23.1756	7.4938
18	15	237	506	307.764	17	18.1038	7.4688
19	10	267	303	244.439	14	17.4599	7.9574
20	15	328	776	1169.86	54	21.6641	8.1675
21	14	312	526	374.635	19	19.7176	7.9321
22	7	176	268	120.125	7	17.1607	7.936
23	17	368	607	475.86	21	22.66	7.4204
24	14	404	623	335.735	17	19.7491	7.9182
25	8	322	608	915.293	35	26.1512	7.1883
26	12	249	609	952.377	38	25.0625	7.5833
27	8	236	400	298.392	15	19.8928	6.9221
28	6	271	653	462.676	18	25.7042	8.3868
29	9	273	695	582.014	24	24.2506	6.8395
30	3	131	267	172.196	9	19.1329	8.6861

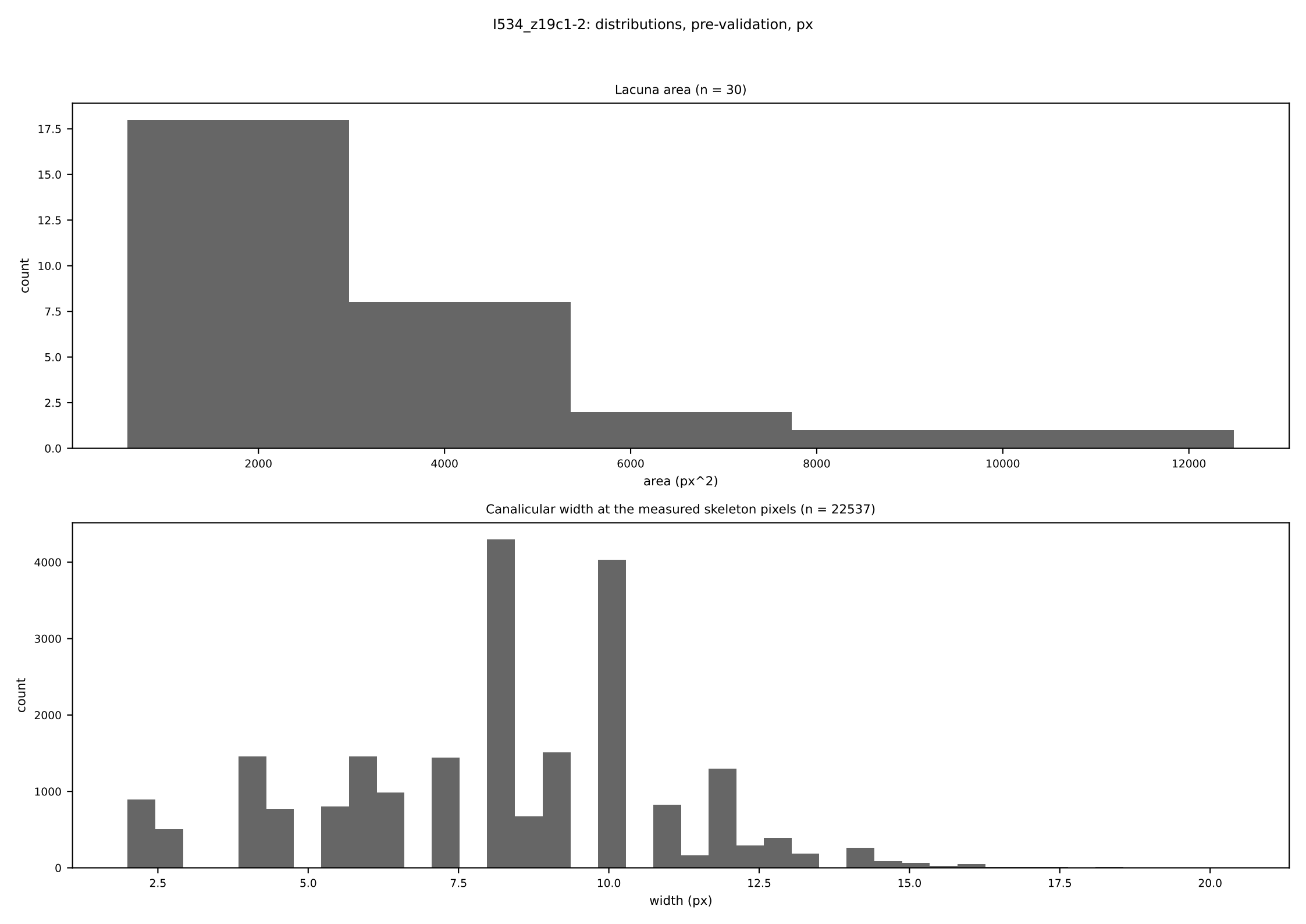
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	685.305	10237	17639	0.6811	0.6379	0.0379994	0.0378139	2.3347
2	195.823	6193	13279	0.7604	0.6834	0.0245438	0.0325326	1.532
3	1095.21	16270	23128	0.64	0.5463	0.0153042	0.0204946	0.6391
4	287.179	8472	15908	0.9855	0.8408	0.019712	0.0215615	2.0893
5	193.823	5534	11776	1	1	0.0424648	0.0363451	4.6434
6	272.936	8089	13798	1	1	0.0216343	0.0276852	1.8319
7	266.788	7353	14089	0.7107	0.6653	0.0142799	0.0136986	1.1245
8	462.008	15283	28880	1	1	0.0317346	0.026385	3.896
9	283.463	8651	18027	1	1	0.0152584	0.0138126	2.4695
10	164.51	7220	19561	1	1	0.0372576	0.0367057	5.4708
11	144.539	6645	18106	1	0.9955	0.020015	0.0214846	3.4593
12	382.854	12834	21714	1	1	0.0297647	0.0271714	5.2239
13	262.871	6200	11714	1	1	0.0385484	0.0404644	3.0433

14	296.191	5917	12004	1	1	0.0562785	0.0519827	4.3891
15	274.735	10367	18090	1	1	0.0230539	0.0182421	3.2759
16	132.539	4341	10609	1	1	0.0562083	0.0403431	7.5449
17	375.103	12127	20489	0.9532	0.8444	0.0347159	0.030309	4.2655
18	300.049	10385	21896	1	1	0.0228214	0.0231092	4.9992
19	155.717	5059	7404	1	1	0.0527772	0.0409238	6.4219
20	288.735	10625	22862	1	1	0.0308706	0.0339428	5.1951
21	439.83	12113	21276	1	1	0.0257575	0.0247227	3.183
22	289.848	8213	14035	1	1	0.0214294	0.0190951	2.4151
23	316.777	10912	23620	1	1	0.0337243	0.0256986	5.3666
24	533.263	15650	25399	1	0.8882	0.0258147	0.0245285	2.6253
25	263.664	9930	21920	1	1	0.032427	0.0277372	3.0342
26	215.723	8071	20618	1	1	0.0308512	0.0295373	5.5627
27	247.279	9530	20421	1	0.9977	0.0247639	0.0195877	3.2352
28	251.22	9124	23312	1	1	0.0297019	0.0280113	2.3883
29	352.617	11633	27463	1	1	0.0234677	0.0253068	2.5523
30	125.154	4302	10972	0.681	0.6229	0.030451	0.0243347	2.397

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	361	548	449.132	762.068	7	8	10
2	115	246	170.125	492.718	4	5	4
3	165	165	275.137	529.86	9	6	7
4	109	215	184.539	384.635	5	6	5
5	212	260	261.924	483.576	4	5	4
6	164	309	201.823	449.588	5	8	8
7	59	101	122.811	220.409	1	1	4
8	419	622	550.943	871.122	14	14	11
9	108	149	146.196	272.38	4	1	4
10	232	469	305.35	816.796	4	9	9
11	99	103	152.439	445.789	2	2	5
12	326	445	428.433	666.5	8	12	10
13	221	384	265.238	525.433	5	8	6
14	306	536	385.433	714.269	7	7	8
15	230	283	276.764	375.777	5	6	3
16	206	403	282.451	495.931	3	7	5
17	409	556	486.931	705.227	10	11	10
18	222	282	265.794	568.973	6	5	4
19	258	264	296.995	333.894	5	8	2
20	304	568	365.007	873.068	7	9	13
21	286	345	346.664	597.6	6	3	5
22	176	256	199.995	306.735	4	3	4
23	317	405	418.647	691.512	8	10	7
24	343	430	460.546	711.583	9	11	10
25	287	439	376.676	700.269	8	9	8
26	249	486	288.836	691.127	5	4	5
27	199	250	270.007	457.517	6	8	7
28	248	512	298.309	724.186	8	7	8
29	238	390	320.706	811.193	7	7	8
30	90	154	149.539	288.037	3	3	5


I534_z19c1-2: distributions, pre-validation, px
Lacuna area (n = 30)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 22537)
count
width (px)